

STUDENT PERFORMANCE PREDICTION AND ANALYSIS USING MACHINE LEARNING

1. PROJECT OVERVIEW

The Student Performance Prediction and Analysis System is a Machine Learning based solution developed to analyze student academic performance and provide valuable educational insights. The system predicts student grades using academic, behavioral, demographic, family, and lifestyle factors. In addition, the system performs dropout risk analysis, personalized learning recommendations, attendance impact analysis, time management analysis, and sentiment analysis on student feedback.

The project aims to assist educational institutions in identifying students who require additional academic support and improve overall learning outcomes through data-driven decision making.

2. OBJECTIVES

The major objectives of the project are:

1. Predict student academic grades using Machine Learning.
2. Analyze factors affecting academic performance.
3. Identify students at risk of poor academic performance.
4. Generate personalized learning recommendations.
5. Study the impact of attendance on grades.
6. Analyze the relationship between study hours and academic performance.
7. Perform sentiment analysis on student feedback.
8. Build a deployable predictive model for educational institutions.

3. DATASETS USED

Multiple educational datasets were collected and evaluated during the project.

Primary Dataset

Students Performance Dataset

Number of Records: 5000

Number of Features: 23

Target Variable:

Grade

Important Features:

- Attendance (%)
- Midterm Score
- Final Score
- Assignments Average
- Quiz Average
- Participation Score
- Project Score
- Study Hours Per Week
- Parent Education Level
- Family Income Level
- Stress Level
- Sleep Hours Per Night
- Internet Access at Home
- Extracurricular Activities

Additional Datasets Used

Student Mathematics Dataset

Features:

- Study Time
- Absences
- Parent Education
- GPA Related Factors

Student Portuguese Dataset

Features:

- Academic Performance Factors
- Attendance

- Study Habits

Teaching Quality Dataset

Features:

- Student Reviews
- Quiz Scores
- Exam Scores
- LMS Activity
- Assignment Submission Rate

Used for Sentiment Analysis.

xAPI Educational Dataset

Used for Educational Data Exploration and Feature Comparison.

4. DATA PREPROCESSING

Data preprocessing was performed to improve data quality and model performance.

Data Cleaning Steps

Step 1: Missing Value Analysis

Missing values were identified in:

- Parent Education Level

Step 2: Missing Value Treatment

Mode Imputation was applied.

Parent Education Level missing values were replaced using the most frequent category.

Step 3: Duplicate Removal

Duplicate records were checked and removed.

Step 4: Removal of Irrelevant Features

The following columns were removed:

- Student_ID
- First_Name
- Last_Name

- Email

Reason:

These attributes do not contribute to academic performance prediction.

Step 5: Text Standardization

Extra spaces and inconsistent formatting were removed.

Step 6: Data Validation

Final dataset was verified for:

- Missing values
- Duplicates
- Data type consistency

5. FEATURE ENGINEERING

The following categories of features were used.

Academic Features

- Midterm Score
- Final Score
- Assignment Average
- Quiz Average
- Participation Score
- Project Score

Behavioral Features

- Attendance Percentage
- Study Hours Per Week

Family Features

- Parent Education Level
- Family Income Level

Lifestyle Features

- Stress Level

- Sleep Hours

Technology Features

- Internet Access at Home

Activity Features

- Extracurricular Activities

6. CATEGORICAL DATA ENCODING

Machine learning algorithms require numerical inputs.

Label Encoding was applied to:

- Gender
- Department
- Parent Education Level
- Family Income Level
- Internet Access
- Extracurricular Activities
- Grade

Encoded values were stored using LabelEncoder and saved for deployment.

7. CLASS IMBALANCE HANDLING

The original grade distribution was highly imbalanced.

Original Distribution:

A = 16

B = 638

C = 2307

D = 1760

F = 279

This imbalance could bias the model toward majority classes.

To solve this issue, SMOTE (Synthetic Minority Oversampling Technique) was applied.

Balanced Distribution:

A = 2307

B = 2307

C = 2307

D = 2307

F = 2307

This ensured equal representation of all grade categories.

8. MACHINE LEARNING MODELS EVALUATED

Several machine learning algorithms were tested and compared.

1. Random Forest Classifier
2. Gradient Boosting Classifier
3. Extra Trees Classifier
4. LightGBM
5. Decision Tree Classifier

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- Cross Validation

9. FINAL MODEL SELECTION

Selected Model:

Random Forest Classifier

Reasons for Selection:

- High prediction accuracy

- Handles mixed feature types efficiently
- Robust against overfitting
- Stable performance during cross-validation
- Good interpretability through feature importance analysis

10. MODEL PERFORMANCE

Final Model:

Random Forest + SMOTE

Performance Metrics:

Test Accuracy:

95.32%

Cross Validation Accuracy:

95.67%

Evaluation Method:

5-Fold Cross Validation

Cross Validation Results:

0.9471

0.9549

0.9545

0.9610

0.9658

Average:

95.67%

11. FEATURE IMPORTANCE ANALYSIS

Feature importance was analyzed using Random Forest.

Most influential features included:

- Midterm Score

- Final Score
- Assignment Average
- Quiz Average
- Attendance
- Study Hours

Feature importance analysis helped identify the factors that most strongly influence student performance.

12. DROPOUT RISK ANALYSIS

A dropout risk analysis module was developed.

Students were categorized into:

- Low Risk
- Medium Risk
- High Risk

Results:

Low Risk = 2584 Students

Medium Risk = 1443 Students

High Risk = 973 Students

Risk categorization was based on:

- Attendance
- Study Hours
- Stress Level

This module helps institutions identify students requiring intervention.

13. PERSONALIZED LEARNING RECOMMENDATIONS

A recommendation engine was implemented.

Recommendations generated include:

- Improve Attendance

- Increase Study Hours
- Maintain Proper Sleep Schedule
- Reduce Stress Through Planning

Recommendations are generated based on individual student behavior and academic indicators.

14. TIME MANAGEMENT ANALYSIS

The relationship between study hours and academic performance was analyzed.

Factors Studied:

- Study Hours Per Week
- Total Academic Score

Scatter plot analysis was performed to visualize the relationship between time invested in learning and academic performance.

15. ATTENDANCE IMPACT ANALYSIS

Attendance was analyzed against academic performance.

Factors Studied:

- Attendance Percentage
- Total Academic Score

This analysis helps understand how classroom participation contributes to student success.

16. SENTIMENT ANALYSIS

Sentiment analysis was performed on student feedback using NLTK.

Tool Used:

NLTK VADER Sentiment Analyzer

Process:

- Review extraction
- Sentiment scoring
- Classification

Results:

Positive Reviews = 1498

Negative Reviews = 602

Positive Feedback Percentage:

71.33%

Negative Feedback Percentage:

28.67%

This analysis provides insights into student satisfaction and teaching effectiveness.

17. MODEL SAVING

The trained models were saved using Joblib.

Generated Files:

student_performance_model.pkl

label_encoders.pkl

academic_model.pkl

These files are used for deployment and prediction.

18. TECHNOLOGIES USED

Programming Language:

Python

Libraries:

- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Imbalanced-learn
- NLTK
- Joblib

Development Environment:

- Visual Studio Code
- Jupyter Notebook

19. PROJECT ACHIEVEMENTS

Successfully completed:

- ✓ Data Collection
- ✓ Data Cleaning
- ✓ Data Preprocessing
- ✓ Feature Engineering
- ✓ Categorical Encoding
- ✓ Class Balancing using SMOTE
- ✓ Multiple Model Evaluation
- ✓ Grade Prediction
- ✓ Dropout Risk Analysis
- ✓ Personalized Recommendations
- ✓ Time Management Analysis
- ✓ Attendance Impact Analysis
- ✓ Sentiment Analysis
- ✓ Model Saving
- ✓ Cross Validation

20. CONCLUSION

The Student Performance Prediction and Analysis System successfully predicts student grades using machine learning techniques and provides valuable educational insights. The Random Forest model trained on a balanced dataset achieved an accuracy of 95.32% and a cross-validation accuracy of 95.67%.

The system further supports dropout risk identification, personalized learning recommendations, attendance impact analysis, time management analysis, and sentiment analysis. These capabilities make the system a useful tool for educational institutions seeking to improve student performance and support data-driven academic decision-making.